

Question 1 (2 points)

Consider the following context-free grammar:

$$S \rightarrow S0S1 \mid S1S0 \mid \epsilon$$

Which of the following is the language generated by this grammar?

- A. Set of all palindromes
- B. Set of all even numbers
- C. Set of all strings having either only 0's or only 1's
- D. Set of all strings having equal number of 0's and 1's**

Option D is the correct answer.

It is clear from the production rules of this grammar that any string generated by this grammar will have equal number of 0's and 1's.

Now we need to show that any string that has equal number of 0's and 1's can be generated by this grammar. We can prove this by induction. Let us assume that all strings of length at most $n - 1$ having equal number of 0's and 1's can be generated using the production rules of this grammar. Let us consider a string x of length n such that it has equal number of 0's and 1's. Let us assume that x ends with a 1 (the other case where x ends with 1 is symmetrical). Now consider the first 0 after 1 in x (counting from right to left). If there is nothing between this 0 and the ending 1, then the remaining string on the left of 0 also must have equal number of 0's and 1's and hence can be generated using the production rules as follows $S \rightarrow S0S1 \rightarrow S0\epsilon 1 \rightarrow S01$. The remaining string can be generated from S (by induction hypothesis). If this is not the case, we try to find the next 0 (moving from right to left) and see if the string between this 0 and the ending 1 has equal number of 0's and 1's. We keep on doing this until we find a 0 which divides x into two substrings p and q (i.e. $x = p0q1$) such that p and q both have equal number of 0's and 1's. We can then use the production rule $S \rightarrow S0S1$ to generate this string. The existence of such a 0 in x is guaranteed because as we move on from right to left, adding more 0's and 1's, the first time the number of 0's and 1's become equal has to be when we see a 0 (because we start with a 1).

Question 2 (2 points)

Consider the following grammar

$$\begin{aligned} S &\rightarrow PQ \mid \epsilon \\ P &\rightarrow PP \mid \epsilon \\ Q &\rightarrow QQ \mid \epsilon \end{aligned}$$

Which of the following regular expressions generates the same language as the given grammar?

- A. 0^*1^***
- B. $0^* + 1^*$
- C. $(0 + 1)^*$
- D. 00^*11^*

There is a typo in this question. It will be reevaluated in due time.

The second and third production rules were supposed to be $P \rightarrow PP \mid 0 \mid \epsilon$ and $Q \rightarrow QQ \mid 1 \mid \epsilon$. Option A i.e. 0^*1^* would be the correct answer then.

Question 3 (2 points)

Consider the following grammar

$$S \longrightarrow 0AB \mid 1BA \mid \epsilon$$

$$A \longrightarrow 0A0 \mid 1A1 \mid \epsilon$$

$$B \longrightarrow 0B1 \mid 1B0 \mid \epsilon$$

Which of the following strings does not belong to the language generated by this grammar?

A. 0011010

B. 0101001

C. 1101000

D. 1011001

Option B is the correct answer.

The variable A generates even-length strings that are palindromes and the variable B generates even-length strings that are anti-palindromes (i.e. none of the corresponding characters match in the string and its reverse). Therefore, S generates strings that are of one of the following forms:

- 1) Begins with 0, an even-length palindrome in the middle, an even-length anti-palindrome at the end.
- 2) Begins with 1, an even-length anti-palindrome in the middle, an even-length palindrome at the end.

Option A can be imagined as $0(0110)(10)$. 0110 is an even-length palindrome and 10 is an even-length anti-palindrome.

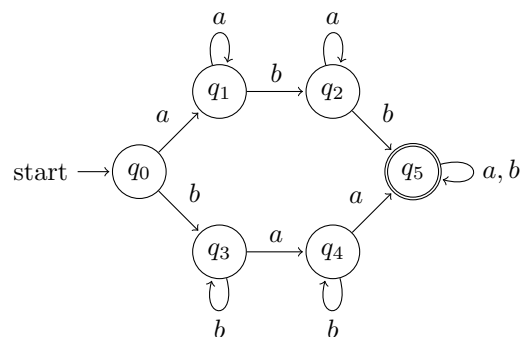
Option C can be imagined as $1(1010)(00)$. 1010 is an even-length anti-palindrome and 00 is an even-length palindrome.

Option D can be imagined as $1(01)(1001)$. 01 is an even-length anti-palindrome and 1001 is an even-length palindrome.

Option B cannot be imagined in any of these forms. This is because the substring 101001 is not an anti-palindrome and does not have a (non-empty) prefix that is a palindrome.

Question 4 (2 points)

Consider the following DFA.



Which of the following context-free grammars generate the same language as the given DFA?

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- A. $S \rightarrow aA \mid bB$
 $A \rightarrow aA \mid bA'$
 $B \rightarrow bB \mid aB'$
 $A' \rightarrow aA' \mid bC$
 $B' \rightarrow bB' \mid aC$
 $C \rightarrow aC \mid bC \mid \epsilon$
- B. $S \rightarrow aA \mid bB$
 $A \rightarrow bA \mid aA'$
 $B \rightarrow aB \mid bB'$
 $A' \rightarrow aA' \mid bC$
 $B' \rightarrow bB' \mid aC$
 $C \rightarrow aC \mid bC \mid \epsilon$
- C. $S \rightarrow aA \mid bB$
 $A \rightarrow aA \mid bA'$
 $B \rightarrow bB \mid aB'$
 $A' \rightarrow aA' \mid bC$
 $B' \rightarrow bB' \mid aC$
 $C \rightarrow aC \mid bC \mid a \mid b$
- D. $S \rightarrow aA \mid bB$
 $A \rightarrow aA \mid bA'$
 $B \rightarrow bB \mid aB'$
 $A' \rightarrow bA' \mid aC$
 $B' \rightarrow aB' \mid bC$
 $C \rightarrow aC \mid bC \mid \epsilon$

Option A is the correct answer.

Using the method given in Section 8.1.4 of Week 4 Lecture notes, we can derive a context-free grammar for the given DFA.

Question 5 (2 points)

Choose the CFG that generates the same language generated by the regular expression $(bab)^*$.

- A. $S \rightarrow babS \mid \epsilon$
B. $S \rightarrow bS \mid aS \mid \epsilon$
C. $S \rightarrow babS \mid bab$
D. $S \rightarrow baS \mid abS \mid \epsilon$

Option A is correct.

Question 6 (2 points)

Which of the following context-free grammars is ambiguous?

- A. $S \rightarrow 0S1 \mid \epsilon$
B. $S \rightarrow 0S1 \mid 0 \mid 1 \mid \epsilon$
C. $S \rightarrow 0S0 \mid 1S1 \mid 0 \mid 1 \mid \epsilon$
D. $S \rightarrow S0 \mid 0 \mid 1 \mid \epsilon$

Option D is correct. Let us consider the string 0. It can be derived in two ways using this grammar:

- 1) $S \rightarrow 0$
- 2) $S \rightarrow S0$
 $\rightarrow 0$ (using $S \rightarrow \epsilon$)

Therefore, this grammar is ambiguous.

Question 7 (2 points)

Let D be a DFA with n states that accepts the language L . Let G be the context-free grammar having the least number of non-terminals that generates L . The number of non-terminals in G is at most (select the smallest possible value)

- A. n
- B. $2n$
- C. n^2
- D. 2^n

Option A is the correct answer.

Refer to Section 8.1.4 of Week 4 Lecture notes. We can construct a grammar with n non-terminals for a DFA with n states. (where each non-terminal in the grammar corresponds to a state in the DFA)

Question 8 (2 points)

Which of the following is the context free grammar that generates the set of all palindromes over the alphabet $\{0, 1\}$?

- A. $S \rightarrow 0S0 \mid 1S1 \mid 0 \mid 1$
- B. $S \rightarrow 0S0 \mid 1S1 \mid 0 \mid 1 \mid \epsilon$
- C. $S \rightarrow 0S0 \mid 1S1 \mid \epsilon$
- D. $S \rightarrow 0S1 \mid 1S0 \mid \epsilon$

Option B is the correct answer. Palindromes are strings which when reversed remains unchanged. Note that for such strings, the last and the first character must be same. Also if we remove the first and last character from the string, the remaining string is also a palindrome. Using this idea, we construct the grammar as given in option B.

Note:- The production rules $S \rightarrow 0$ and $S \rightarrow 1$ are used to generate odd-length palindromes (eg.- 101) where there is a single middle character. On the other hand, the production rule $S \rightarrow \epsilon$ is used to generate even-length palindromes which don't have a single middle character.